

Logo	Ministry of Higher Education and Scientific Research
	Directorate General for University Renovation

Descriptive sheet of a Teaching Unit (UE) and its components (ECUE)

Module Sheet: Fundamentals of Natural Language Processing

Teaching Unit (UE)	Data Science Foundations	UE Code	UEF210	ECUE	Fundamentals of NLP
Programme	Mathematics	Department: Applied Mathematics		Specialization: Data Science	

Type of teaching	Lectures	24 h	Regime	Continuous Assessment (mixed)	
	Tutorials (TD)	18 h		Mixed	X
	Practical labs (TP)	18 h		Semester	S2
	Project	0 h			

Total volume (per semester)	In-class: 60 h	Coefficient	2.5
	Estimated personal workload: 30 h	ECTS credits	5

Module coordinator	Aymen Ben Brik	Lecturer	Aymen Ben Brik
Target audience	M1 GAMMA & L2 LMAD — Data Science	Contributors	—

Prerequisites	• Basic Python programming (functions, lists, dictionaries, file I/O). • Foundations of machine learning (supervised learning, train/test split, basic classifiers). • Linear algebra: vectors, matrices, dot products, ℓ_2 norm, distances. • Elementary probability and statistics (distributions, conditional probability).
----------------------	---

Intended learning outcomes (ILO)	By the end of this module, the student will be able to: • ILO1: Define NLP and identify its core challenges (ambiguity, context, variability). • ILO2: Recognize the major NLP tasks and frame real-world problems in the appropriate task category. • ILO3: Situate NLP within the broader AI / ML / DL landscape and identify the dominant learning paradigms. • ILO4: Distinguish discriminative from generative modelling and apply Bayes' rule to bridge the two. • ILO5: Trace the historical evolution of NLP and explain the technical motivations behind each era. • ILO6: Apply tokenization strategies (word, character, sub-word/BPE) appropriate to a given language and task. • ILO7: Build sparse (BoW, TF-IDF) and dense (Word2Vec, GloVe, FastText) text representations. • ILO8: Compute and interpret cosine, dot-product and Euclidean similarities for retrieval and clustering. • ILO9: Evaluate NLP systems with task-appropriate metrics (Precision, Recall, F_1, perplexity, BLEU, ROUGE) and recognise their pitfalls.
--	---

Content	Chapter 1 — NLP Landscape (12 h) • Section 1: What is NLP? Definition, scope, core challenges • Section 2: Major NLP tasks and applications (taxonomy by I/O structure) • Section 3: NLP within the AI ecosystem; supervised, unsupervised, self-supervised • Section 4: Discriminative vs. generative models; Bayes bridge • Section 5: Historical evolution: rules → statistics → embeddings → Transformers Chapter 2 — Text Representation and Evaluation (12 h) • Section 1: Tokenization fundamentals (word, character, BPE/WordPiece/SentencePiece) • Section 2: Sparse representations: Bag-of-Words, TF-IDF, document-term matrix • Section 3: Dense representations: distributional hypothesis, Word2Vec (skip-gram & CBOW), GloVe, FastText, vector arithmetic, biases and limitations • Section 4: Semantic similarity: cosine, dot product, Euclidean; the retrieval pattern; ANN libraries • Section 5: Evaluation metrics: classification (Precision, Recall, F_1, ROC/PR), generation (perplexity, BLEU, ROUGE) Practical labs (18 h) • Lab 1: Tokenization (whitespace, regex, NLTK, BPE from scratch, Hugging Face tokenizers) • Lab 2: BoW + TF-IDF text classifier on 20 Newsgroups • Lab 3: Word2Vec embeddings on Brown corpus + analogies + PCA visualization • Lab 4: Evaluation metrics (confusion matrix, ROC/PR, perplexity, BLEU, ROUGE)
Teaching and learning methods	The pedagogical framework relies on constructive and active learning, with a strong experiential component through practical labs. In-class Mixed (X) Distance

Teaching techniques	Active learning • Flipped classroom: short pre-class videos and notes, in-class problems and code reviews. • Live coding sessions in Jupyter notebooks (NLTK, scikit-learn, gensim, Hugging Face). • Pair programming during lab sessions; collaborative debugging of NLP pipelines. Constructive learning • Examples and counter-examples for ambiguity, polysemy, embedding analogies. • Progressive construction of artefacts: BPE algorithm, classifier, retrieval index. • Guided → autonomous: scaffolding is reduced over the semester. Experiential learning • Hands-on manipulation of real datasets (20 Newsgroups, Brown corpus, code-switched text). • Empirical comparison of model families on the same task. • Failure analysis: tokenizer mismatches on dialectal Arabic, class-imbalance pitfall. Reflective learning • Self-assessment grids on code quality, scientific clarity, and reproducibility. • Peer review of lab notebooks. • Written justification of design choices (which tokenizer, which similarity, which metric).
Assessment methods	Assessment combines formative (feedback-driven) and summative (graded) components, aligned with the intended learning outcomes.

Assessment	Associated activity	ILOs assessed	Type
Diagnostic	Quiz on Python, basic ML, linear algebra	ILO1, ILO3	Ungraded
Formative 1	Tokenization exercises and short BPE walkthrough	ILO1, ILO6	Feedback
Formative 2	Building a TF-IDF text classifier and inspecting features	ILO2, ILO7, ILO9	Constructive
Formative 3	Training Word2Vec; nearest neighbours and analogy tests	ILO5, ILO7	Experiential
Continuous Assessment (CC)	Mini-project: end-to-end NLP pipeline (tokenize → embed → classify → evaluate) on a chosen dataset	ILO1 → ILO9	Summative
Final exam	Written exam: theory questions, short calculations (TF-IDF, cosine, F_1 , BLEU), failure-mode analysis	ILO1 → ILO9	Summative

Bibliography & resources	Reference textbooks • Jurafsky, D., & Martin, J.H. <i>Speech and Language Processing</i> (3rd ed., draft). • Eisenstein, J. <i>Introduction to Natural Language Processing</i>, MIT Press, 2019. • Manning, C.D., Raghavan, P., & Schütze, H. <i>Introduction to Information Retrieval</i>, Cambridge UP, 2008. Foundational papers • Mikolov et al. (2013), <i>Efficient estimation of word representations in vector space</i> (Word2Vec). • Pennington, Socher, & Manning (2014), <i>GloVe</i>. • Bojanowski et al. (2017), <i>Enriching word vectors with subword information</i> (FastText). • Vaswani et al. (2017), <i>Attention is all you need</i>. • Devlin et al. (2018), <i>BERT</i>. Tools and tutorials • NLTK: https://www.nltk.org/ • scikit-learn: https://scikit-learn.org/ • gensim Word2Vec tutorial: https://radimrehurek.com/gensim/ • Hugging Face: https://huggingface.co/ Datasets used • 20 Newsgroups (sklearn built-in) • Brown corpus (NLTK) • Sample Tunisian dialect snippets (provided in lab notebooks)
Course repository	https://github.com/Aymenbenbrik/FundamentalsNLP
Document version	v1.0 — compiled May 6, 2026